ERIC HERNANDEZ

eric-hdez@proton.me | (562) 940-7822 | linkedin.com/in/eric-hdez | eric-hdez.dev | U.S. Citizen

SUMMARY

Software engineer with 2.5+ years building high-throughput data systems at Bloomberg, from databases and caching through client-facing UIs. Backend-focused, full-stack capable. Day-to-day in C++ and Python.

EXPERIENCE

Bloomberg LP | **Software Engineer** Mar 2024 – Present

- Cut total cluster CPU 71% within weeks on an overloaded production database by moving 5 hot queries onto an auto-invalidating cache, with a C++ accessor library and feature-flag rollout
- Migrated an entity-data service (∼4B requests/day) to a new database: re-architected caching by moving JSON cache entries to protobuf in one merged Redis cache, and added Kafka-driven invalidation
- Designed and shipped a benchmark-lookup microservice end to end (design doc, APIs, Redis-backed caching, C++ client library; 0.76 ms latency), replacing a capacity-limited lookup that blocked some rate-schedule refreshes
- Owned and built a client-facing G10 cross-currency money-market matrix end to end (UI, plus a backend that recomputes and publishes to subscribers every 30s)
- Fixed a memory leak that repeatedly crashed a client-facing Terminal screen (15 merged incident reports)
- Consolidated writes from multiple offline tasks into one real-time listener on an internal data pipe, centralizing the write path and improving data consistency
- Migrated 3 legacy C/C++ production systems from Solaris to Linux, validated side by side, zero downtime
- Built Grafana dashboards for 188 databases and 50+ microservices, used by leadership across 4 teams

UC Santa Cruz ITS | **Student Software Developer** Jan 2023 – Sep 2023

- Improved a poster-ordering web app (upload, submit, pay) for 16,000+ students and faculty: enhanced form validation and formatting so clean data reached the backend, and strengthened payment error handling

NVIDIA Corporation | **Software Engineer Intern** Jun 2021 – Aug 2021

- Built an automated Perforce code-review framework for the engineering team; debugged a GPU resource manager on Linux, helping resolve a security bug

SKILLS

Languages: C++, C, Python, SQL, TypeScript, JavaScript
Backend & Systems: Schema-first RPC and REST APIs, Microservices, Caching, Event-driven systems
Datastores & Messaging: PostgreSQL, Redis, Kafka, Comdb2
Infrastructure & Observability: Docker, Linux, Git, Grafana, pytest, GoogleTest

EDUCATION

Georgia Institute of Technology | **M.S. Computer Science (OMSCS)** Aug 2026 – Expected 2029
Online, part-time

University of California Santa Cruz | **B.S. Computer Science** Sep 2019 – Jun 2023
GPA: 3.89. Relevant Coursework: Compiler Design (Graduate Level), Computer System Design, Full Stack Web Development I, II, Software Design Principles

ORGANIZATIONS

Bloomberg LP, Fixed Income and Derivatives Engineering | Engineering Representative
Society of Hispanic Professional Engineers, Professional Chapter | Member